Roll No.: | | | | | | | | | | | |

Test Date: 19-01-2025

Aakash
Medical | IIT-JEE | Foundations

E
Booklet Code

Aakash Delhi/NCR Zone

AIATS
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2025

National Eligibility-cum-Entrance Test (NEET)

TEST No. 4

(XII Passed Students)

Einstein X369

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.
9. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
10. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
11. There are two sections in each subject i.e., Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions out of 15 from Section-B.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 4

TOPICS OF THE TEST

Physics	Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves
Chemistry	Solutions, Chemical Kinetics, Organic Chemistry and Some Basic Principles and Techniques
Botany	Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development
Zoology	Chemical Coordination and Integration, Animal Kingdom, Structural Organisation in Animals– Animal Morphology (Cockroach and Frog)

Einstein X369
Medical/IIT-JEE Foundations
(AR-001)

MM : 720

TEST - 4

Time : 3 Hrs. 20 Min.

[PHYSICS]

Choose the correct answer :

SECTION-A

1. If thermal conductivity is represented by A and that of thermal resistance by B, then dimensions of $A^{-1}B^{-1}$ will be

   (1) [L] (2) [L^2]

   (3) [LT] (4) [LT^{-1}]

2. If a hot body cools from 85°C to 75°C in time t_1, from 75°C to 65°C in time t_2 and from 65°C to 55°C in time t_3, then the correct relation between t_1, t_2 and t_3 will be (temperature of surrounding = 30°C)

   (1) $t_1 = t_2 = t_3$ (2) $t_1 > t_2 > t_3$

   (3) $t_1 < t_2 < t_3$ (4) $t_1 < t_2 > t_3$

3. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

   Assertion (A): When a gas is allowed to expand in a piston cylinder arrangement, work done by gas is positive.

   Reason (R): In expansion of a gas the force due to gaseous pressure and displacement (of piston) are in the opposite direction.

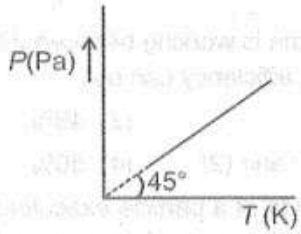
   (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

   (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

   (3) (A) is correct but (R) is incorrect.

   (4) (A) is incorrect but (R) is correct.

4. Figure shows pressure (P) vs temperature (T) variation for 3 moles of an ideal gas. The volume of the gas is nearly equal to (R = 8.314 J K^{-1} mol^{-1})



   (1) 2 m^3

   (2) 4 m^3

   (3) 25 m^3

   (4) 16 m^3

5. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

   Assertion (A): Carnot engine is a reversible engine operating between two temperatures T_1 (source) and T_2 (sink).

   Reason (R): The Carnot engine consists of three adiabatic and one isothermal process.

   (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

   (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

   (3) (A) is correct but (R) is incorrect.

   (4) (A) is incorrect but (R) is correct.

Space for Rough Work

6. For a given amount of an ideal gas, if X be the slope of pressure-volume curve for its isothermal process and Y be the slope of pressure-volume curve for its adiabatic process, then $\frac{C_P}{C_V}$ is equal to

(1) $\frac{X}{Y}$ (2) $\frac{Y}{X}$

(3) XY (4) $\frac{X^2}{Y}$

7. A heat engine is working between 27°C and 327°C. Its possible efficiency can be

(1) 40% (2) 48%

(3) Both (1) and (2) (4) 60%

8. The amplitude of a particle executing SHM is made three-fourth keeping its time period constant. If initial total energy of particle is E, then its total energy after amplitude is made three-fourth will be

(1) $\frac{E}{2}$ (2) E

(3) $\frac{3}{4}E$ (4) $\frac{9}{16}E$

9. The temperature of a body on Celsius scale is found to be x°C. When it is measured by Fahrenheit thermometer, it is found to be x°F, then the value of x is

(1) –100 (2) –40

(3) –50 (4) –300

10. A harmonic progressive wave having frequency 600 Hz is travelling with a speed of 360 m/s. The minimum separation between two points which are 60° out of phase will be

(1) 0.12 m (2) 0.10 m

(3) 0.8 m (4) 0.6 m

11. If the temperature of a gas sample is halved keeping pressure constant, then the mean free path of the gas molecule becomes

(1) Double

(2) Half

(3) One-fourth

(4) Four times

12. Ratio of average kinetic energy of a molecule of an ideal monoatomic gas to average kinetic energy per molecule of an ideal triatomic non-linear gas at room temperature will be

(1) 1 : 1 (2) 1 : 2

(3) 2 : 1 (4) 1 : 3

13. If Q is net heat supplied to an ideal monoatomic gas at constant pressure, then work done by the gas will be

(1) $\frac{2}{5}Q$ (2) $\frac{1}{5}Q$

(3) $\frac{3}{5}Q$ (4) $\frac{4}{5}Q$

14. In which of the following processes, temperature of the system decreases?

(1) Isobaric expansion

(2) Adiabatic expansion

(3) Adiabatic compression

(4) Isothermal expansion

15. The displacement of a particle in SHM varies according to the relation $x = 4(\cos\pi t + \sin\pi t)$. The equivalent relation is

(1) $4\sqrt{2}\sin\left(\pi t + \frac{\pi}{4}\right)$ (2) $4\sqrt{2}\cos\left(\pi t + \frac{\pi}{4}\right)$

(3) $4\sin\left(\pi t + \frac{\pi}{4}\right)$ (4) $4\cos\left(\pi t + \frac{\pi}{4}\right)$

Space for Rough Work

16. A gas is allowed to expand, work done by gas will be maximum in which of the following thermodynamic process?
(1) Isothermal (2) Isobaric
(3) Adiabatic (4) Isometric

17. Match the following for work done in various thermodynamic processes. (Symbols have their usual meaning)

List-I (Process)		List-II (W.D.)	
A.	Isobaric expansion	(I)	Zero
B.	Isochoric	(II)	$-\Delta U$
C.	Adiabatic	(III)	ΔQ
D.	Isothermal	(IV)	$nR\Delta T$

(1) A-IV, B-I, C-II, D-III (2) A-IV, B-I, C-III, D-II
(3) A-IV, B-II, C-III, D-I (4) A-I, B-II, C-III, D-IV

18. Select the correct option about stationary wave.
(1) In stationary wave, two particles must differ in phase by zero radian.
(2) In stationary wave, nodes being permanently at rest, so no energy can be transmitted across them.
(3) In stationary wave, two particles differ in phase by $\frac{\pi}{2}$ radian.
(4) Both (1) & (2)

19. The equation for a simple harmonic plane progressive wave propagating along the positive direction of x-axis is (Symbols have their usual meanings)
(1) $y = A\sin(\omega t + kx)$ (2) $y = A\sin k(vt + x)$
(3) $y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$ (4) $y = A\sin 2\pi\left(\frac{t}{T} + \frac{2x}{\lambda}\right)$

20. A particle is executing SHM according to the equation $x = A\sin\omega t$, then the distance from the mean position where the kinetic energy of particle equals the potential energy, will be nearly equal to
(1) 50% of A (2) 85% of A
(3) 70% of A (4) 65% of A

21. If temperature of an ideal gas is increased by 300%, then root mean square speed of the gas will become
(1) Half (2) Twice
(3) Thrice (4) Quadrupled

22. A monoatomic gas undergoes a polytropic process. The molar specific heat capacity of the gas for this process is R. Equation of polytropic process will be given by (All symbols have their usual meaning)
(1) PV^{-3} = constant
(2) PV^{3} = constant
(3) PV^{-2} = constant
(4) PV^{2} = constant

23. Match the List I and List II. (Symbols have their usual meanings)

	List I		List II
a.	v_{rms}	(i)	$\frac{3}{2}k_BT$
b.	Pressure (P)	(ii)	nC_vT
c.	Translational K.E. per molecule	(iii)	$\sqrt{\frac{3RT}{M}}$
d.	U (Internal energy)	(iv)	$\frac{1}{3}\rho v_{rms}^2$

(1) a(iii), b(iv), c(ii), d(i)
(2) a(iii), b(iv), c(i), d(ii)
(3) a(iii), b(ii), c(i), d(iv)
(4) a(iii), b(i), c(iv), d(ii)

Space for Rough Work

24. 2 moles of oxygen is mixed with 4 moles of Helium then the specific heat of the mixture at constant volume is (All symbols have their usual meaning)

(1) $\frac{7}{6}R$ (2) $\frac{11}{6}R$

(3) $\frac{13}{6}R$ (4) $\frac{5}{6}R$

25. If average kinetic energy of a molecule of hydrogen gas at 100°C is E, then average kinetic energy of a molecule of oxygen at 200°C is

(1) E (2) $2E$

(3) $\frac{E}{2}$ (4) $1.27E$

26. When two travelling waves of same amplitude, frequency and velocity moving in opposite directions are superposed, in general the phenomenon observed is

(1) Echo (2) Standing waves

(3) Beats (4) Diffraction

27. In a Carnot engine, efficiency is 40% at hot reservoir temperature T. For efficiency to be 50%, hot reservoir temperature should be (assume temperature of sink constant)

(1) $\frac{T}{5}$ (2) $\frac{3T}{2}$

(3) $\frac{T}{2}$ (4) $\frac{6T}{5}$

28. Two spheres are made of same substance and have radius in the ratio 1 : 2. Their thermal capacities are in the ratio of

(1) 1 : 2

(2) 1 : 4

(3) 1 : 8

(4) 2 : 1

29. A black body has maximum intensity wavelength λ_m at temperature 2000 K. Its corresponding wavelength at temperature 3000 K will be

(1) $\frac{1}{3}\lambda_m$ (2) $\frac{1}{9}\lambda_m$

(3) $\frac{2}{3}\lambda_m$ (4) $\frac{4}{9}\lambda_m$

30. Read the following statements and select the correct option.

Statement I: If an ideal gas container in motion is suddenly stopped, the temperature of the gas rises.

Statement II: The temperature of an ideal gas is a measure of the average kinetic energy of a molecule.

In the light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct while statement II is incorrect

(4) Statement I is incorrect while statement II is correct

31. A particle is performing SHM on a straight line with time period of 12 seconds. The ratio of minimum time taken to maximum time taken by particle to cover a distance equal to amplitude will be

(1) $\frac{1}{1}$ (2) $\frac{1}{2}$

(3) $\frac{2}{3}$ (4) $\frac{3}{2}$

32. The displacement x of a particle executing simple harmonic motion is given as $x = 4\sin(\pi + 2t)$ m. Maximum speed of the particle is

(1) 4 m/s (2) 2 m/s

(3) 8 m/s (4) 8π m/s

Space for Rough Work

33. In a resonance tube experiment the first resonance is obtained for 10 cm of air column. If end correction for this apparatus is 0.5 cm, then second resonance is obtained for
(1) 31 cm (2) 30 cm
(3) 11 cm (4) 15 cm

34. Water at 60°C is mixed with equal amount of ice at 0°C. The final temperature of the mixture will be
(1) 30°C (2) 10°C
(3) 0°C (4) 1°C

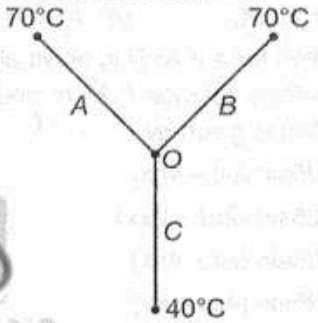
35. Read the following statements and select the correct option.
Statement I: For displacement antinode in a longitudinal stationary wave in air, there is minimum change in pressure and density.
Statement II: Position of displacement antinode is same as that of pressure node.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

SECTION-B

36. The third overtone produced by a vibrating string (fixed at both ends) 2 m long is 600 Hz. Fundamental frequency of vibration is
(1) 300 Hz (2) 150 Hz
(3) 1200 Hz (4) 200 Hz

37. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In simple harmonic motion, the phase difference between displacement and acceleration is π.
Reason (R): In simple harmonic motion, the graph between velocity and the displacement is an ellipse.
In the light of the above statements, the correct option is
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

38. Three identical conductors A, B and C are connected at point O as shown in figure. The temperature of the junction O is



(1) 60°C (2) 50°C
(3) 55°C (4) 45°C

39. Equation of a standing wave is given by
(1) $y = A\sin(\omega t - kx)$
(2) $y = A\sin\omega t$
(3) $y = 2A\sin kx \cos\omega t$
(4) All of the above

40. A particle of mass 200 g executes SHM. The restoring force is provided by a spring of force constant 80 N/m. The time period of oscillation is
(1) 0.314 s (2) 0.628 s
(3) 0.5 s (4) 0.75 s

Space for Rough Work

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41. A body is executing simple harmonic motion with an angular frequency of 2 rad/s. The velocity of the body at 2 cm displacement, when amplitude of motion is 6 cm, is
(1) 11.3 cm/s (2) 5 cm/s
(3) 4 cm/s (4) 10.4 cm/s

42. Two metal rods A and B of lengths in ratio 2 : 1 have same temperature difference between their ends. Their thermal conductivities are K_1 and K_2 and cross-sectional area A_1 and A_2 respectively. If the rate of heat conduction in A is four times that in B, then
(1) $4K_2A_2 = K_1A_1$ (2) $8K_2A_2 = K_1A_1$
(3) $K_2A_1 = 4K_1A_2$ (4) $K_2A_1 = 2A_2K_1$

43. The equation for a travelling wave along negative x direction with amplitude 0.25 m and wavelength λ equals 0.5 m is given by
(1) $y = 0.25\sin(\omega t - 4\pi x)$
(2) $y = 0.25\sin(2\omega t + 2\pi x)$
(3) $y = 0.25\sin(\omega t + 4\pi x)$
(4) $y = 0.25\sin(4\omega t - kx)$

44. The equation of state $PV = \frac{RT}{2}$ corresponds to (where symbols have their usual meaning)
(1) 16 g of O_2 (2) 1 g of H_2
(3) 24 g of N_2 (4) Both (1) and (2)

45. Time period of simple pendulum having length as 1 m is 2 seconds. If length of pendulum is decreased by $\frac{7}{16}$ m, then new time period will be
(1) 1.5 seconds
(2) 2.5 seconds
(3) 2 seconds
(4) 1 second

46. A particle is executing SHM with time period T. If it starts its motion from mean position, then its velocity at $\frac{T}{12}$ second will be
(Where $T = \frac{2\pi}{\omega}$ and A is amplitude of the particle)
(1) $\frac{A\omega}{4}$ (2) $\frac{\sqrt{3}A\omega}{2}$
(3) $\frac{A\omega}{2}$ (4) $\frac{\sqrt{3}A\omega}{4}$

47. Sound waves transfer
(1) Only energy
(2) Only momentum
(3) Both energy and momentum
(4) Only mass

48. A ball of metal has a concentric spherical cavity within it. If the ball is heated, the volume of the cavity will
(1) Increase
(2) Decrease
(3) Remain same
(4) First increase then decrease

49. Molar specific heat at constant volume for rigid diatomic gas is
(1) $\frac{7}{2}R$ (2) $\frac{5}{2}R$
(3) $\frac{5}{3}R$ (4) $\frac{7}{5}R$

50. Work done by an ideal gas during adiabatic process in expanding from state (P_1, V_1) to state (P_2, V_2) is given by
(1) $\frac{P_1V_1 - P_2V_2}{\gamma - 1}$ (2) $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$
(3) $\frac{P_1V_1 - P_2V_2}{1 + \gamma}$ (4) $\frac{P_2V_2 - P_1V_1}{1 + \gamma}$

Space for Rough Work

[CHEMISTRY]

SECTION-A

51. Aqueous solutions were prepared by dissolving 20 g of glucose ($C_6H_{12}O_6$) in 500 g of water (P_1), 20 g of urea (CH_4N_2O) in 500 g of water (P_2) and 20 g of sucrose ($C_{12}H_{22}O_{11}$) in 500 g of water (P_3). The right option for the decreasing order of freezing point of these solutions is

(1) $P_2 > P_1 > P_3$ (2) $P_3 > P_1 > P_2$

(3) $P_3 > P_2 > P_1$ (4) $P_1 > P_2 > P_3$

52. The mixture which shows negative deviation from Raoult's law is

(1) Benzene + Toluene

(2) Chloroethane + Bromoethane

(3) Acetone + Chloroform

(4) Ethanol + Acetone

53. An ideal solution is made by mixing A and B. If partial vapour pressure of component A in the solution is 250 torr. Then molar ratio of A to B in vapour phase is (P_A^o = 500 torr, P_B^o = 200)

(1) $\frac{2}{5}$ (2) $\frac{5}{2}$

(3) $\frac{5}{7}$ (4) $\frac{2}{7}$

54. Which of the following concentration terms is/are temperature independent?

(I) Molarity (II) Normality

(III) % (w/w) (IV) Molality

(1) (I) and (II) only

(2) (II) and (III) only

(3) (III) and (IV) only

(4) (IV) only

55. K_H values for some gases at the same pressure 'P' and temperature 'T' are given

	Gas	K_H/(kbar)
(a)	Carbon dioxide	1.67
(b)	Argon	40.3
(c)	Methane	0.413
(d)	Vinyl chloride	0.611

Where K_H is Henry's Law constant in water. The order of their solubility in water is

(1) b < a < c < d (2) b < a < d < c

(3) a < b < d < c (4) b < d < a < c

56. Given below are two statements.

Statement I: Aquatic species are more comfortable in warm water than in cold water.

Statement II: The solubility of gases in liquid decreases with decrease in temperature.

In the light of above statements choose the correct option.

(1) Statement I is correct and statement II is incorrect

(2) Statement I is incorrect and statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

57. Which one will exhibit the largest elevation of boiling point? (Assume 100% dissociation of electrolyte)

(1) 0.10 m KCl (aq)

(2) 0.20 m $C_6H_{12}O_6$(aq)

(3) 0.10 m $Al_2(SO_4)_3$(aq)

(4) 0.20 m K_2SO_4(aq)

Space for Rough Work

58. Given below are two statements, one is labelled as Assertion (A) and other is Reason (R)

Assertion (A): Azeotropic mixture cannot be separated by fractional distillation.

Reason (R): Nitric acid and water can form maximum boiling azeotrope.

In the light of above statement, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

59. If molality of the dilute solution is halved, the value of molal depression constant (K_f) will be

(1) Doubled (2) Halved

(3) Tripled (4) Unchanged

60. What is the mole fraction of the solute in a 2.00 m aqueous solution?

(1) 2.067 (2) 0.052

(3) 1.034 (4) 0.035

61. 400 mL of an aqueous solution of a protein contains its 2.52 g. The osmotic pressure of this solution at 300 K is found to be 2.57×10^{-3} bar. The molar mass of protein will be ($R = 0.083$ L bar mol^{-1} K^{-1})

(1) 32011 g mol^{-1} (2) 51038 g mol^{-1}

(3) 61038 g mol^{-1} (4) 71052 g mol^{-1}

62. The following data were obtained during the first order thermal decomposition of a gas A at constant volume, $A(g) \rightarrow 2B(g) + C(g)$

S.No.	Time/s	Total pressure/(atm)
1.	0	0.1
2.	115	0.28

The rate constant of the reaction is

(1) 2×10^{-3} sec^{-1} (2) 4×10^{-4} sec^{-1}

(3) 2×10^{-2} sec^{-1} (4) 3×10^{-6} sec^{-1}

63. Mechanism of a hypothetical reaction $2A_3 \rightarrow 3A_2$ is given below.

(a) $A_3 \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} A_2 + A$

(b) $A_3 + A \xrightarrow{\text{slow}} 2A_2$

The overall order of the reaction will be

(1) 0 (2) 1

(3) 2 (4) 3

64. If initial concentration of the reactant is halved, the half-life period of a zero order reaction

(1) Is halved

(2) Is doubled

(3) Becomes one-fourth

(4) Remains unchanged

65. In the reaction

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(\ell).$$

The ratio of rate of appearance of water (H_2O) to the rate of disappearance of Bromide ions (Br^-) is

(1) 3 : 5

(2) 5 : 3

(3) 1 : 1

(4) 1 : 3

66. If the rate constant for a first order reaction is k then the time required for 99.99% completion of the reaction is

(1) $\frac{2.303}{k}$ (2) $\frac{3 \times 2.303}{k}$

(3) $\frac{4 \times 2.303}{k}$ (4) $\frac{10 \times 2.303}{k}$

Space for Rough Work

67. Consider the following statements.
(a) Order of a reaction is an experimental quantity.
(b) Order is applicable only to elementary reactions.
(c) Molecularity cannot be zero.
Select the correct statements.
(1) (a) and (b) only (2) (a) and (c) only
(3) (b) and (c) only (4) (a), (b) and (c)

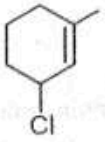
68. In a reaction A + B → Product, rate of reaction is doubled when the concentration of A is doubled and the rate becomes eight times when the concentration of both the reactants (A and B) are doubled, rate law for the reaction can be written as
(1) Rate = $k[A][B]$ (2) Rate = $k[A]^2[B]$
(3) Rate = $k[A][B]^2$ (4) Rate = $k[A]^0[B]^2$

69. The graph between ln k versus $\frac{1}{T}$ is a straight line.
The ratio of slope to intercept of the line is
(1) $\frac{-E_a}{R \times \ln A}$ (2) $\frac{-R \times \ln A}{E_a}$
(3) $\frac{-E_a \times \ln A}{R}$ (4) $-\frac{R}{E_a \times \ln A}$

70. Consider the following statements regarding to the equation, Rate = $PZ_{AB}\, e^{-E_a/RT}$
(a) Z_{AB} represents the collision frequency of reactants, A and B.
(b) $e^{-E_a/RT}$ represents the fraction of molecules with energies less than E_a.
(c) P takes into account the fact that in a collision, molecules must be properly oriented.
Select the correct statements.
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

71. The number of sp^2 hybridised carbon atoms and π bonds in 2-methylpent-1-en-4-yne respectively are
(1) 3, 2 (2) 2, 3
(3) 3, 4 (4) 4, 3

72. The IUPAC name of the compound



is
(1) 3-Chloro-1-methylcyclohex-1-ene
(2) 1-Chloro-3-methylcyclohex-2-ene
(3) 1-Methyl-3-chlorocyclohex-1-ene
(4) 2-Chloro-5-methylcyclohex-1-ene

73. The most stable carbocation, among the following is
(1) $CH_3 - \overset{\oplus}{C}H - CH_2 - CH_3$
(2) $CH_3 - \overset{\oplus}{C}(CH_3)_2 - CH_3$
(3) $CH_3 - C(CH_3)_2 - \overset{\oplus}{C}H - CH_3$
(4) $CH_3 - \overset{\oplus}{C}H_2$

74. Given below are two statements.
Statement I: Thin layer chromatography is a type of adsorption chromatography.
Statement II: Paper chromatography is a type of partition chromatography.
In the light of above statements, choose the correct option.
(1) Statement I is false but Statement II is true
(2) Both statement I and statement II are true
(3) Both statement I and statement II are false
(4) Statement I is true but statement II is false

Space for Rough Work

75. Aniline is separated from a mixture of water and aniline by using
(1) Sublimation
(2) Crystalization
(3) Differential extraction
(4) Steam distillation

76. Consider the following statements.
(a) $Fe_4[Fe(CN)_6]_3 \cdot xH_2O$ is Prussian blue in colour.
(b) Phosphorus in an organic compound cannot be detected by 'Lassaigne's test'.
(c) Carbon and hydrogen are detected by heating the compounds with cupric oxide.
The correct statements are
(1) (a) and (b) only (2) (a), (b) and (c)
(3) (a) and (c) only (4) (b) and (c) only

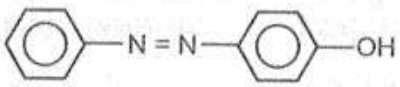
77. Match the **column I** with **column II** and select the correct option.

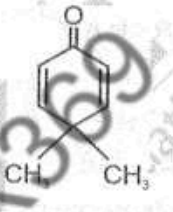
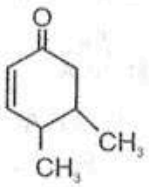
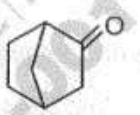
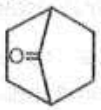
	Column I (Compound)		**Column II (Common Name)**
a.	$C_6H_5OCH_3$	(i)	Acetophenone
b.	$C_6H_5COCH_3$	(ii)	Aniline
c.	$CH_3OCH_2CH_3$	(iii)	Anisole
d.	$C_6H_5NH_2$	(iv)	Ethyl methyl ether

(1) a(iii), b(i), c(ii), d(iv)
(2) a(iii), b(i), c(iv), d(ii)
(3) a(i), b(iii), c(iv), d(ii)
(4) a(i), b(iv), c(iii), d(ii)

78. Correct order of -I effect of the given species is
(1) –OH > – F > – CN
(2) –F > – OH > – CN
(3) –CN > – OH > – F
(4) –CN > – F > – OH

79. $\overset{\ominus}{C}H_2 - \overset{O}{\overset{\|}{C}} - CH_2 - CH_3$ and $CH_2 = \overset{:\ddot{O}:^{\ominus}}{\overset{|}{C}} - CH_2 - CH_3$ are
(1) Resonating structures
(2) Tautomers
(3) Geometrical isomers
(4) Optical isomers

80. Kjeldahl's method for estimation of nitrogen is applicable for which of the given compounds?
(1) $CH_3 - CH_2 - NH_2$
(2) $CH_3 - CH_2 - NO_2$
(3) 
(4) 

81. Which among the given molecules can exhibit tautomerism?

(a)

(b)

(c)

(d)
(1) (b) and (c) only (2) (c) and (d) only
(3) (a) and (b) only (4) (a) and (d) only

82. The total number of structural isomers possible with the molecular formula $C_4H_{10}O$ is
(1) 4 (2) 5
(3) 7 (4) 3

Space for Rough Work

EinsteinX369 Medical/IIT-JEE (AR-01)

83. In Duma's method for estimation of nitrogen, 0.30 g of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm pressure. If the aqueous tension at 300 K is 15 mm, then the percentage of nitrogen in the compound is
(1) 20.95
(2) 17.25
(3) 24.90
(4) 18.32

84. Consider the following statements regarding the nucleophile.
(a) Nucleophiles attack at low electron density sites
(b) Nucleophiles are electron seeking species
(c) Ammonia is a nucleophile
Select the correct statements.
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

85. The decomposition of gaseous ammonia on a hot platinum surface at high pressure is an example of
(1) Zero order reaction
(2) First order reaction
(3) Pseudo first order reaction
(4) Second order reaction

SECTION-B

86. Which of the following colligative property is widely used to determine molar masses of proteins and polymers?
(1) Relative lowering in vapour pressure
(2) Osmotic pressure
(3) Depression in freezing point
(4) Elevation in boiling point

87. A 6.84% (w/V) solution of cane sugar ($C_{12}H_{22}O_{11}$) is isotonic with 2% (w/V) solution of a substance X. The molecular weight of X is (X is non-volatile non-electrolyte)
(1) 68.4 u (2) 170.3 u
(3) 100 u (4) 101.4 u

88. Increasing order of priority of functional group according to the IUPAC nomenclature is
(1) $-COOH < -OH < -NH_2 < -COCl$
(2) $-NH_2 < -OH < -COOH < -COCl$
(3) $-COOH < -COCl < -OH < -NH_2$
(4) $-NH_2 < -OH < -COCl < -COOH$

89. A solution of urea boils at 101.024°C at the atmospheric pressure. If k_f and k_b for water are 1.86 and 0.512 K kg mol^{-1} respectively, the above solution will freeze at
(1) –1.86°C (2) –0.512°C
(3) –3.72°C (4) –1.024°C

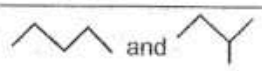
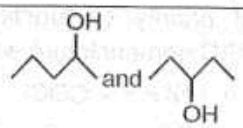
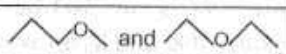
90. A solute undergoes 30% trimerization in the solution. The van't Hoff factor of the solute is
(1) 0.70 (2) 0.80
(3) 0.30 (4) 0.40

91. The mass of non-volatile solute (molar mass 60 g mol^{-1}) which should be dissolved in 171 g of octane to reduce its vapour pressure to 60% is
(1) 60 g (2) 75 g
(3) 100 g (4) 90 g

92. If for a reaction, rate constant becomes double with every 10°C rise in temperature then the ratio of rate of reaction at 25°C to 55°C is
(1) $\frac{1}{4}$ (2) 4
(3) $\frac{1}{8}$ (4) 8

Space for Rough Work

93. Match the **column I** with **column II**.

	Column I (Structures)		**Column II (Type of isomers)**
a.		(i)	Metamers
b.		(ii)	Functional group isomers
c.	CH_3-CH_2-CHO and $CH_3-\overset{\overset{O}{\|\|}}{C}-CH_3$	(iii)	Chain isomers
d.		(iv)	Position isomers

Select the correct option.

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(ii), d(i)

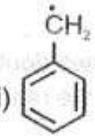
(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iv), d(iii)

94. The correct order of stability of the given radicals is

(i) $CH_3-\dot{C}H_2$

(ii) $CH_2=CH-\dot{C}H_2$

(iii) 

(1) (i) > (ii) > (iii)

(2) (i) > (iii) > (ii)

(3) (iii) > (ii) > (i)

(4) (ii) > (iii) > (i)

95. Non-benzenoid compound among the following is

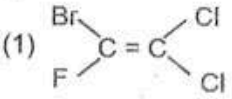
(1) Naphthalene

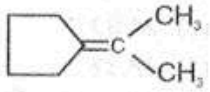
(2) Aniline

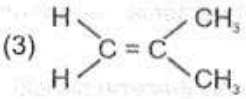
(3) Benzene

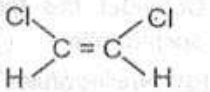
(4) Tropone

96. Which of the following compounds can show geometrical isomerism?

(1) 

(2) 

(3) 

(4) 

97. In Carius method of estimation of Bromine 1 g of an organic compound gave 1.128 g of AgBr. The percentage of bromine in the compound is

[Molar mass of AgBr = 188 g mol^{-1}, gram atomic mass of Br = 80 g mol^{-1}]

(1) 75% (2) 48%

(3) 35% (4) 26%

98. The Lassaigne's extract is boiled with conc. HNO_3 while testing for halogens. By doing so it

(1) Helps in the precipitation of AgCl

(2) Decomposes Na_2S and NaCN, if formed

(3) Decreases the solubility product of AgCl

(4) Decreases the concentration of NO_3^- ions

99. For a reaction, activation energy E_a = 14 kJ and the rate constant at 400 K is $1.6 \times 10^8\ s^{-1}$. The rate constant at 500 K will be

[Given that gas constant, R = 8.314 J K^{-1} mol^{-1} and log 2.32 = 0.365]

(1) $1.6 \times 10^6\ s^{-1}$ (2) $3.7 \times 10^8\ s^{-1}$

(3) $3.7 \times 10^2\ s^{-1}$ (4) $3.2 \times 10^4\ s^{-1}$

Space for Rough Work

100. Consider the following statements.
(a) Inhibitor reduces the rate of reaction
(b) The catalyst provides an alternate pathway or reaction mechanism by reducing the activation energy between reactants and products
(c) Catalyst catalyses the spontaneous as well as non-spontaneous reactions

Select the correct statements.
(1) (a) and (b) only (2) (a) and (c) only
(3) (b) and (c) only (4) (a), (b) and (c)

[BOTANY]

SECTION-A

101. In dicots, the meristems which are responsible for increase in girth of the plant is/are
(1) Root apical meristems
(2) Shoot apical meristems
(3) Intercalary meristems
(4) Cork cambium

102. Which of the following features are present in typical meristematic cells?
a. Dense protoplasm
b. Inconspicuous nucleus
c. Very large vacuole
d. Primary cell wall
e. Plasmodesmatal connections
f. Secondary cell wall
(1) b, d and e only (2) c and d only
(3) a, b and f only (4) a, d and e only

103. In protoplasmic type of respiration, which of the following respiratory substrates is used?
(1) Fat (2) Carbohydrate
(3) Protein (4) Glycerol

104. Read the following and find the **correct** statements w.r.t. growth pattern that show(s) a typical S-shaped curve.
a. This type of growth exemplified by elongation of root at a constant rate.
b. The stationary phase of growth is achieved due to the shortage of space, food and accumulation of toxins.
c. This kind of growth is observed when zygote develops into embryo.

Select the **correct** answer from the following options.
(1) a and c only (2) b only
(3) a and b only (4) b and c only

105. Who amongst the following performed the experiment through which first action spectrum of photosynthesis was described?
(1) Ruben (2) T.W. Engelmann
(3) Hatch and Slack (4) Blackman

106. What kind of growth curve is observed by microorganisms, when they are provided with limited food and space?
(1) Linear (2) Sigmoid
(3) Parabolic (4) Hyperbolic

107. How many redox-equivalents are removed from one molecule of PGAL during the following reaction?
3-Phosphoglyceraldehyde (PGAL) → 1, 3-bisphosphoglycerate
(1) Six
(2) Two
(3) Eight
(4) Four

Space for Rough Work

108. How many of the following is/are the important intrinsic factor(s), which influence(s) the developmental processes of the plant?

Genetic, Water, Nutrition, Oxygen, PGRs, Temperature

(1) 2 (2) 3
(3) 4 (4) 1

109. Match the given columns and select the **correct** option.

	Column I		Column II
a.	Joseph Priestley	(i)	Release of O_2 from green parts of plants in the presence of sunlight
b.	Jan Ingenhousz	(ii)	Bell jar experiments
c.	Julius von Sachs	(iii)	Water is the source of oxygen
d.	Cornelius van Niel	(iv)	Glucose is produced in green parts of plants during photosynthesis

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iii), c(iv), d(i)
(3) a(ii), b(i), c(iv), d(iii) (4) a(ii), b(i), c(iii), d(iv)

110. The maximum number of ATP produced, if fructose-1,6-bisphosphate is used as the substrate for aerobic respiration is
(1) 38 ATP (2) 36 ATP
(3) 40 ATP (4) 42 ATP

111. The compound phosphoenol pyruvate is the primary CO_2 acceptor in which of the following plants?
(1) C_2 plants (2) C_3 plants
(3) C_4 plants (4) Both C_3 and C_4 plants

112. In which of the following processes, decarboxylation occurs?
a. Glycolysis
b. Krebs' cycle
c. Link reaction
d. Alcoholic fermentation
e. Lactic acid fermentation
(1) a, b and d only
(2) b, c and d only
(3) c and e only
(4) d and e only

113. Match List-I with List-II.

	List-I		List-II
a.	Citrate synthase	(i)	Oxidative phosphorylation
b.	Aldolase	(ii)	Link reaction
c.	ATP synthase	(iii)	EMP pathway
d.	Pyruvate dehydrogenase	(iv)	TCA cycle

Choose the **correct** answer from the options given below.
(1) a(iii), b(ii), c(iv), d(i)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(iv), b(iii), c(i), d(ii)
(4) a(iv), b(i), c(iii), d(ii)

114. Which of the following phytohormones was isolated from a fungus which causes the foolish seedling disease in rice plant?
(1) Auxins
(2) Gibberellins
(3) Cytokinins
(4) Ethylene

Space for Rough Work

115. The components/features that are the part of non-cyclic photophosphorylation reaction but not of cyclic photophosphorylation reaction are
a. PS I
b. Splitting of water
c. $NADP^+$ synthesis
d. Water as electron donor
e. Reaction centre (P_{680})
Choose the **correct** options from the following
(1) a, c and e only (2) b, d and e only
(3) a, b and c only (4) b, c and e only

116. In plants, light saturation is seen at ____ of the total sunlight available to the plants.
Select the **correct** option to fill in the blank.
(1) 10% (2) 20%
(3) 35% (4) 50%

117. Which of the following equations can **correctly** represents the exponential growth?
(1) $\frac{W_1}{W_0} = rt$ (2) $\frac{W_0}{W_1} = e \times rt$
(3) $\frac{W_1}{W_0} = e^{rt}$ (4) $W_0 = W_1 + e^{rt}$

118. What is ATP : NADPH ratio required in the reduction phase of Calvin cycle for 2 molecules of CO_2 fixed?
(1) 2 : 1 (2) 1 : 1
(3) 1 : 2 (4) 3 : 1

119. What is the value of R.Q. when tripalmitin is used as a substrate of respiration?
(1) 0.7 (2) 1
(3) 0.9 (4) 2

120. Given below are the two statements, one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Chlorophyll *a* is considered to be primary photosynthetic pigment associated with photosynthesis.
Reason (R): Action spectrum of photosynthesis corresponds to the absorption spectrum of chlorophyll *a*.
In the light of the above statements, choose the **correct** answer from the given options.
(1) (A) is true but (R) is false
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(4) Both (A) and (R) are false

121. The plant hormone, which is used to increase the number of female flowers in cucumber, is also used for
(1) Stimulating closure of stomata
(2) Inducing bolting in rosette plants
(3) Initiating germination of seeds, such as peanuts
(4) Inducing apical dominance

122. Which of the following pairs of hormones have antagonistic effect w.r.t. seed germination in plants?
(1) Ethylene and auxins
(2) Auxins and gibberellins
(3) Cytokinins and auxin
(4) Gibberellins and ABA

123. Identify the **odd** one w.r.t. cyclic photophosphorylation from the following.
(1) An external source of electron is not required
(2) It synthesises ATP only
(3) It does not have reaction centre
(4) It occurs in the stroma lamellae

124. The first, 6-carbon compound which is formed in the mitochondrial matrix during the Krebs' cycle is
(1) α-ketoglutaric acid (2) Citric acid
(3) Fumarate (4) Oxalosuccinic acid

Space for Rough Work

125. Read the statements given below.

Statement I: The phenomena where certain living differentiated cells regain the ability to divide and form new cells is called dedifferentiation.

Statement II: Secondary xylem and cork are the examples of redifferentiated tissues.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both the statements I and II are incorrect
(2) Both the statements I and II are correct
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

126. Find the **incorrect** statement w.r.t. chloroplast.

(1) It has a stack of thylakoids called the granum.
(2) Adjacent grana are connected by stroma lamellae.
(3) Pigments that function in photochemical reactions of photosynthesis are embedded in the inner membrane of chloroplast envelope.
(4) Dark reactions of photosynthesis, where there is an enzymatic synthesis of sugars, take place in stroma.

127. How many substrate level phosphorylation reactions will occur during Krebs' cycle for one glucose molecule?

(1) 2 (2) 6
(3) 4 (4) 8

128. Which of the following plant growth regulators, when applied to the cut end of decapitated shoots, prevents the growth of lateral buds into branches?

(1) GA_3 (2) ABA
(3) Kinetin (4) IBA

129. Which of the following is a cold sensitive enzyme involved in the C_4 cycle?

(1) RuBisCO (2) ATP synthase
(3) PEP synthetase (4) NADP reductase

130. Which of the following are **true** for the shield pigments associated with photosynthesis?

a. They form the main photosynthetic pigment in plants.
b. They absorb incoming light of different wavelengths so as to enable wider range of wavelength for photosynthesis.
c. They form the reaction centre.
d. Some of them appear yellowish-orange in the chromatogram.
e. They prevent photo-oxidation of chlorophyll *a* in plants.

Choose the **correct** ones.

(1) a, c and e only
(2) b, c and d only
(3) a, b, d and e only
(4) b, d and e only

131. The PGRs which were discovered as kinetin are involved in

(1) Speeding up the malting process in brewing industry
(2) Morphogenesis in tissue culture
(3) Promoting abscission of flowers and fruits
(4) Sprouting of storage organs like rhizomes and tubers

132. Treatment of a plant with cytokinins will enhance

(1) Ageing of leaves
(2) Apical dominance
(3) Apical hook formation in dicot seedlings
(4) Chloroplast production in leaves

Space for Rough Work

133. Which plant hormone was discovered as a volatile substance, emitted by ripened oranges that brought about early ripening of bananas when kept nearby?
(1) Auxins (2) Kinetin
(3) Ethylene (4) ABA

134. Which enzyme of the electron transport system is associated with the conversion step of the Krebs' cycle where $FADH_2$ is formed?
(1) Pyruvate dehydrogenase
(2) Succinate dehydrogenase
(3) Lactate dehydrogenase
(4) NADH dehydrogenase

135. Which of the following plants is known to show heterophylly due to change in environment?
(1) Cotton (2) Coriander
(3) Larkspur (4) Buttercup

SECTION-B

136. How many ATPs are formed from substrate level phosphorylation when a molecule of phosphoenol pyruvate is used as a substrate for aerobic respiration?
(1) 4 (2) 2
(3) 3 (4) 5

137. Read the following assumptions and select the **correct** ones w.r.t. the assumptions made for respiratory balance sheet for glucose.
a. There is a sequential, orderly pathway functioning, with one substrate forming the next.
b. Some of the intermediates are utilized to synthesize other compounds.
c. The NADH synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation.
d. Only glucose is being respired and no other alternative substrates are entering in the pathway.
Choose the **correct** answer from the following options.
(1) a and b only (2) a, c and d only
(3) a and c only (4) c and d only

138. Identify the **correct** role of ethylene from the following.
(1) Quicker maturity in conifers
(2) Mobilization of nutrients
(3) Thining of fruits like cotton, cherry and walnut
(4) Elongation of fruits

139. In Z-scheme of photophosphorylation, which of the following is the source of electrons, that are used to replace the electrons lost from the reaction centre of PS II?
(1) Water (2) PS I
(3) Oxygen (4) Carbon dioxide

140. Maximum absorption of light by chlorophyll *b* is seen in the range of wavelength
(1) 400-500 nm (2) 500-600 nm
(3) 300-400 nm (4) 650-750 nm

141. RQ value is more than unity for
(1) Carbohydrate (2) Fats
(3) Proteins (4) Organic acids

142. For photosynthesis, double fixation of carbon dioxide does **not** occur in
(1) Sugarcane (2) *Amaranthus*
(3) *Sorghum* (4) Bell pepper

143. Which of the following pairs of hormones has synergistic effect on abscission?
(1) Gibberellin and Cytokinin
(2) Ethylene and ABA
(3) Abscisic acid and Cytokinins
(4) Cytokinin and Ethylene

Space for Rough Work

144. Elongation of stems in deep water rice plants is promoted by
(1) Auxins (2) Ethylene
(3) Kinetin (4) Gibberellins

145. Match List-I with List-II.

List-I	List-II
a. Tryptophan	(i) Ethylene
b. Acetyl CoA	(ii) Auxins
c. Methionine	(iii) Gibberellins
d. Violaxanthin	(iv) ABA

Choose the **correct** answer from the options given below.
(1) a(iv), b(i), c(iii), d(ii) (2) a(ii), b(iii), c(i), d(iv)
(3) a(ii), b(iii), c(iv), d(i) (4) a(iv), b(ii), c(i), d(iii)

146. Read the following statements and select the **incorrect** statement regarding C_4 plants.
(1) They can tolerate higher temperatures and they lack the wasteful process called photorespiration
(2) These plants are adapted to dry tropical regions
(3) Bundle sheath cells in the leaves of these plants are characterized by large number of chloroplasts, thin wall and no intercellular spaces
(4) They have a special anatomy of leaves called 'Kranz' anatomy

147. The scientist who isolated auxin from the tips of coleoptiles of oat seedlings was
(1) F.W. Went
(2) Charles Darwin and Francis Darwin
(3) F. Skoog
(4) E. Kurosawa

148. Assimilation number shows a relationship between
(1) Chlorophyll and water
(2) Chlorophyll and photosynthesis
(3) Oxygen and chlorophyll
(4) Chlorophyll and CO_2 evolution

149. State **True (T)** or **False (F)** for the given statements and select the **correct** option.
(a) Auxin is responsible for initiation and promotion of cell division and controlling xylem differentiation.
(b) ABA initiates protein and RNA synthesis so as to promote senescence of leaves.
(c) The first natural cytokinin was obtained from autoclaved herring sperm DNA.
(d) Ethylene decreases sensitivity of the roots to gravity.

	(a)	(b)	(c)	(d)
(1)	T	F	T	T
(2)	T	F	F	T
(3)	T	T	F	F
(4)	F	T	F	T

150. Identify the **odd** one for abscisic acid.
(1) ABA induce the dormancy of buds and seeds
(2) Synthesis of ABA is stimulated in adverse environmental conditions and it is called the "stress hormone"
(3) ABA acts as an antitranspirant and causes closure of stomata
(4) Inhibitor-B, abscission II and dormins, are three chemically different inhibitors, collectively called ABA

Space for Rough Work

EinsteinX369 Medical/IIT-JEE Foundations (AIR 001)

[ZOOLOGY]

SECTION-A

151. All of the following options are true for the organisms belonging to the phylum Porifera, **except**
(1) Cells lining the spongocoel and canals are choanocytes
(2) Water flows from ostia to spongocoel from where it goes out *via* osculum
(3) Presence of water vascular system
(4) Body's skeleton is made up of spicules and spongin fibres

152. Which of the following organisms exhibits bilateral symmetry, metamerism and is dioecious?
(1) *Pheretima* (2) *Nereis*
(3) *Hirudinaria* (4) *Balanoglossus*

153. Ventral heart and a dorsally located hollow CNS are present in
(1) *Aplysia* (2) *Fasciola*
(3) *Anopheles* (4) *Balaenoptera*

154. First and second pair of wings in *Periplaneta americana* differ from each other in the fact that latter
(1) Are used in flight
(2) Arise from prothoracic region
(3) Are opaque, dark and leathery
(4) Are known as tegmina

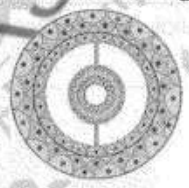
155. In each segment of *Periplaneta americana*, exoskeleton has hardened plates that are joined to each other by
(1) Sclerites
(2) Thick and rigid articular membrane
(3) Arthrodial membrane
(4) Tergites

156. An undifferentiated layer, mesoglea is present between germ layers in
(1) *Aplysia*, *Chaetopleura*
(2) *Physalia*, *Antedon*
(3) *Dentalium*, *Adamsia*
(4) *Gorgonia*, *Pleurobrachia*

157. Select the **correct** option w.r.t. organisms that are pseudocoelomates.
(1) Malpighian tubules are present for excretion
(2) Presence of an excretory tube which removes body wastes through excretory pore
(3) Possess statocyst
(4) Body is covered by a calcareous shell

158. All of the following animals exhibit radial symmetry, **except**
(1) Adult stage of *Meandrina*
(2) Larval stage of *Asterias*
(3) Adult stage of *Ophiura*
(4) Larval stage of *Ctenoplana*

159. Observe the figure given below.



Select the **incorrect** option w.r.t. this figure.
(1) Platyhelminths have the above type of coelomic arrangement
(2) It represents true coelom which is lined by mesoderm
(3) *Ancylostoma* does not exhibit this type of condition
(4) Arthropods have this kind of coelomic cavity

EinsteinX3669

Space for Rough Work

160. In humans, the hormones that are synthesized by hypothalamus and transported axonally to neurohypophysis are
(1) GnRH and somatostatin
(2) TSH and ACTH
(3) Oxytocin and ADH
(4) LH and oxytocin

161. **Assertion (A):** Hypersecretion of PTH can cause weakening of the bones in humans.
Reason (R): PTH acts on the bones and stimulates the process of bone resorption.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are false
(3) (A) is true but (R) is false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

162. How many of the hormones given in the box below interact with membrane bound receptors and generate second messengers?

FSH, Secretin, Insulin, Estradiol, Testosterone, Vasopressin, CCK

Choose the **correct** option.
(1) Six
(2) Five
(3) Four
(4) Three

163. Select the **correct** match w.r.t. humans.
(1) Acromegaly – Hyposecretion of GH in middle age
(2) Diabetes mellitus – Prolonged hyperglycemia and ketonuria
(3) Addison's disease – Underproduction of hormones secreted by adrenal medulla
(4) Exopthalmic goitre – A form of hypothyroidism

164. Notochord is persistent throughout the life in
(1) *Ascidia* and *Branchiostoma*
(2) *Salpa* and *Scoliodon*
(3) *Carcharodon* and *Branchiostoma*
(4) *Doliolum* and *Pristis*

165. Cartilaginous fishes differ from bony fishes in the way that former have
(1) To swim constantly to avoid sinking
(2) 4 pairs of gills which are covered by an operculum on each side
(3) Cycloid scales on skin
(4) An air bladder which regulates buoyancy

166. In frogs, all of the following sense organs are cellular aggregations around the nerve endings, **except**
(1) Sensory papillae
(2) Nasal epithelium
(3) Taste buds
(4) Internal ears

167. Select the **correct** match w.r.t. CNS of frog.
(1) Forebrain – Olfactory lobes and paired diencephalon
(2) Midbrain – Characterised by a pair of optic lobes
(3) Hindbrain – Consists of medulla, pons and cerebellum
(4) Foramen magnum – Pons passes out through it to continue into spinal cord

Space for Rough Work

EinsteinX369 Medical/IIT-JEE Foundation (AR-ECT)

168. Select the **incorrect** option w.r.t. cyclostomes.
(1) Possess an open circulatory system
(2) Larvae present in freshwater but adult forms are marine
(3) They have a sucking and circular mouth
(4) They have 6-15 pairs of gill slits for respiration

169. Match column I with column II w.r.t. organisms and their characteristics.

	Column I		Column II
a.	Bioluminescence	(i)	*Obelia*
b.	Collar cells	(ii)	Sponges
c.	Metagenesis	(iii)	Molluscs
d.	Visceral hump and muscular foot	(iv)	Ctenophores

Choose the **correct** option.
(1) a(iv), b(iii), c(ii), d(i)
(2) a(iv), b(ii), c(i), d(iii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(iv), b(ii), c(iii), d(i)

170. In humans, hormone 'X' released from the hypothalamus inhibits the release of GH from anterior pituitary gland.
Identify the hormone representing 'X' and choose the **correct** option.
(1) GnRH (2) Somatostatin
(3) PRL (4) FSH

171. Read the following statements.
Statement A: Hormones which interact with nuclear receptors mostly regulate gene expression by the interaction of hormone-receptor complex with the genome.
Statement B: Thyroxine and cortisol are steroidal hormones.
Select the **correct** option.
(1) Only statement A is correct
(2) Both statements A and B are correct
(3) Only statement B is correct
(4) Both statements A and B are incorrect

172. Malpighian tubules in cockroach are
(1) Also known as proventriculus
(2) Present at the junction of foregut and midgut
(3) 6-8 in number
(4) 100-150 yellow coloured thin filaments

173. Select the **correct** statement w.r.t. cockroaches.
(1) Females produce 14-16 oothecae, each containing 9-10 eggs.
(2) Six ganglia lie in the thorax.
(3) Spermatophores are formed in males.
(4) The next to last nymphal stage have wings.

174. In frogs, bile from gall bladder and pancreatic juice from the pancreas pass through _____ into duodenum.
Select the **correct** option to fill in the blank.
(1) Hepatic portal system
(2) Renal portal system
(3) A common bile duct
(4) A hepatic duct

175. How many of the structures given in the box below directly open into/passes through cloaca in a male frog?

Urinary bladder, Urinogenital duct, Rectum, Vasa efferentia, Fat bodies, Testes

Select the **correct** option.
(1) Four (2) Three
(3) Two (4) Five

Space for Rough Work

176. All of the following animals possess open circulatory system, **except**
(1) *Locusta*
(2) *Saccoglossus*
(3) *Periplaneta*
(4) *Rana*

177. Match column I with column II.

	Column I		Column II
a.	*Antedon*	(i)	Platypus
b.	*Dentalium*	(ii)	Wall lizard
c.	*Ornithorhynchus*	(iii)	Sea lily
d.	*Hemidactylus*	(iv)	Tusk shell

Select the **correct** option.
(1) a(iii), b(i), c(ii), d(iv)
(2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(i), b(ii), c(iii), d(iv)

178. In the blood report of a person suffering from simple goitre, increased plasma levels of which of the following hormones is most likely to be reported?
(1) Melatonin
(2) ACTH
(3) TSH
(4) Triiodothyronine

179. Decrease in the secretion of a steroidal hormone from growing ovarian follicles can
(1) Increase the bone resorption in females
(2) Decrease the anti-inflammatory reactions in the body
(3) Stimulate the maturation of epididymis in males
(4) Increase the pitch of voice in females

180. The phylum which consists of exclusively marine animals is
(1) Platyhelminthes
(2) Ctenophora
(3) Mollusca
(4) Porifera

181. All of the following hormones increases the blood glucose levels, **except**
(1) Hormones released from neurohypophysis
(2) Hormones released from adrenal medulla
(3) The corticoids, which are released from zona fasciculata
(4) Hormone released from α-cells of pancreas

182. In *Pila*, feather-like gills are present
(1) In the space between mantle and shell
(2) Within file-like rasping organ
(3) In the space between the visceral hump and the mantle
(4) In the anterior head region

183. In *Asterias*, digestive system is __X__ with mouth on __Y__ side of the body.
Identify X and Y and select the **correct** option.

	X	Y
(1)	Complete	Ventral
(2)	Incomplete	Lower
(3)	Complete	Dorsal
(4)	Incomplete	Dorsal

184. In which of the following animals, males and females can be differentiated morphologically?
(1) Tapeworms
(2) Liver flukes
(3) Roundworms
(4) Earthworms

Space for Rough Work

185. Select the **incorrect** match w.r.t. organisms and the excretory structure present in them.
(1) *Periplaneta* – Protonephridia
(2) *Anopheles* – Malpighian tubule
(3) *Balanoglossus* – Proboscis gland
(4) *Nereis* – Nephridia

SECTION-B

186. Choose the **incorrect** option w.r.t. melatonin.
(1) Influences menstrual cycle
(2) Helps in maintaining the body temperature
(3) Influences metabolism
(4) Exclusively regulates hyperpigmentation of skin

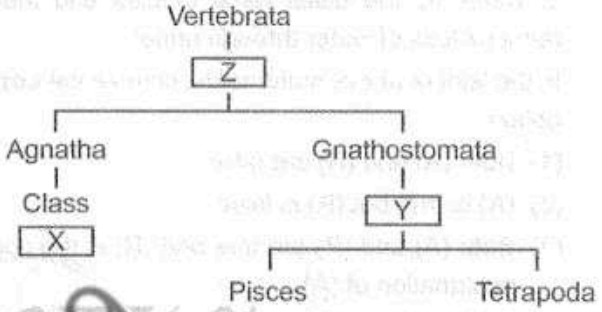
187. Select the **correct** match.
(1) Tissue level of body organisation – *Sycon*
(2) Incomplete digestive tract – *Planaria*
(3) A viviparous aschelminth – *Enterobius vermicularis*
(4) Cells are arranged as loose cell aggregates – *Pennatula*

188. Which of the following statements is **true** w.r.t. humans?
(1) In females, LH induces ovulation of follicles and degrades the corpus luteum.
(2) Pars intermedia secretes only one hormone called MSH.
(3) ACTH stimulates the synthesis and secretion of glucocorticoids from the adrenal medulla.
(4) LH is a gonadotropin which is synthesised in female gonads

189. The steroid hormone that stimulates the formation of alveoli in mammary gland in human females is
(1) Progesterone (2) Estrogen
(3) Prolactin (4) Oxytocin

190. Select the feature which is common between the birds and cockroaches.
(1) Skin is without glands even at the base of tail
(2) Presence of two pairs of wings
(3) Presence of crop and gizzard
(4) Cartilaginous endoskeleton

191. Observe the given flowchart.



Identify Z, X and Y and choose the **correct** option.

	Z	X	Y
(1)	Division	Protochordata	Class
(2)	Division	Cyclostomata	Super-class
(3)	Phylum	Urochordata	Division
(4)	Sub phylum	Cyclostomata	Genus

192. In the heart of a frog, a chamber that pumps out mixed blood, opens into _____ which is present on the ventral side of the heart.
Select the **correct** option to fill in the blank.
(1) Conus arteriosus
(2) Sinus venosus
(3) Left atrium
(4) Right atrium

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EinsteinX369 Medical IIT-JEE Foundation (Aakash)

193. Choose the **correct** statement w.r.t. *Rana tigrina*.
(1) The colour of dorsal side of the body is uniformly pale yellow.
(2) Body is divided into head, trunk and tail.
(3) The hindlimbs end in five webbed digits and are larger and muscular than forelimbs.
(4) Heart is neurogenic.

194. **Assertion (A):** Vasopressin is also known as anti-diuretic hormone.
Reason (R): Vasopressin stimulates reabsorption of water by the distal renal tubules and thereby reduces loss of water through urine.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are false
(2) (A) is true but (R) is false
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

195. Select the **correct** option w.r.t. thymosin.
(1) It is a steroidal hormone
(2) Promotes production of antibodies which provide cell-mediated immunity
(3) Facilitates the differentiation of T-lymphocytes which mainly provide humoral immunity
(4) Source gland of this hormone is located between lungs, behind sternum, on the ventral side of aorta

196. All of the following hormones work antagonistically to each other, **except**
(1) TCT and PTH w.r.t. blood calcium level
(2) Glucagon and insulin w.r.t. blood glucose level
(3) Epinephrine and norepinephrine w.r.t. cardiac output
(4) Aldosterone and ANF w.r.t. blood pressure

197. Match column I with column II.

	Column I		**Column II**
(a)	Gastrin	(i)	Acts on the exocrine pancreas
(b)	Secretin	(ii)	Stimulates the secretion of pepsinogen
(c)	CCK	(iii)	Neuropeptides
(d)	Endorphins	(iv)	Stimulates the release of bile juice from gall bladder

Select the **correct** option.
(1) a(iv), b(iii), c(ii), d(i)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(ii), b(iv), c(iii), d(i)

198. Choose the **incorrect** match w.r.t. cockroaches.

(1)	A pair of testes	–	Laterally in the 4th to 6th abdominal segments
(2)	Mushroom-shaped gland	–	6th to 7th abdominal segments
(3)	Ovaries	–	Laterally in the 2nd to 6th abdominal segments
(4)	A pair of spermatheca	–	Present in 4th segment

Space for Rough Work

199. Read the following statements.

Statement A: Each compound eye present at the dorsal surface of the head in cockroach consists of about 2000 hexagonal ommatidia.

Statement B: Cockroach has mosaic vision with more sensitivity but less resolution.

Select the **correct** option.

(1) Both statements A and B are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

200. Which of the following features is not **true** for cockroach?

(1) Meroblastic determinate cleavage during embryonic development

(2) Abdomen has ten segments

(3) Mandible and labium are unpaired structures

(4) Tracheae open outside through spiracles

❑ ❑ ❑

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